Simulation 3: Possible Sources of Error

Where the model may differ from the real Walkerville network, and how each point could be checked

Ashis Jose Kalathil | Optimal Sensor Placement for Wastewater Network Monitoring, Capstone team PG-S2-22

1. Purpose

Simulation 3 grows one area of Walkerville at a time and asks which manholes would notice it. This report lists the places where the model is simpler than the real network, so the team knows which results to trust and what to ask SA Water for. Every point below was checked against the Simulation 3 code, its settings file and its saved runs on the team repository (main branch, 5 Nov 2026). Where the direction of an error is not known, the report says so rather than guessing.

Rating. **High**: could change which manholes are chosen. **Medium**: could change depths or timing, but probably not the overall pattern. **Low**: small, or affects only a few manholes.

2. How Simulation 3 works, in brief

- EPA SWMM (dynamic wave) on 887 council pipes and 328 manholes draining to node 606, where the council data ends.
- Each run holds every inflow constant until all levels stop moving (up to 1,800 minutes). The result is one settled depth per manhole.
- Inflow has three parts. At the Design wet case it is about 100 L/s in total: sewage 33.4 L/s, infiltration and inflow (I&I) 41.2 L/s, and inflow from pipes outside the council data 25.5 L/s.
- Growth: one of eight areas (R1 to R8) gets new homes on a build-out ladder (10% to 300% of its share of the 485-home PlanSA high case), in eight weather and leakage cases, 448 runs in all.
- A manhole “sees” growth if its depth rises by 10, 25 or 50 mm, newly passes the 150 mm alarm, or newly surcharges.

3. What is not a problem

The numerical side of the runs is healthy, so the points below are about what the model assumes, not about the solver.

- All 448 growth runs and all 8 baselines settled, except one Dry run that reached the ceiling. The largest drift was 0.31 mm.
- The SWMM continuity error was at most 0.47% in any run.
- There are no adverse or flat pipes. Inverts and flow directions come from the published records.

4. Possible errors

A. Network data

No.	What the model does	Why it may differ from reality	Rating	How to check
A1	Manhole depth (ground level minus floor) is taken from ground contours for 322 of 328 manholes; only 5 have a surveyed lid level, and 1 has none and is given 0.5 m.	Contours can be off by a few tenths of a metre. This changes the spill level and the “half shaft” level. It does not change surcharge or fullness, which use the published pipe inverts and diameters.	**Medium**	SA Water lid levels for the manholes chosen as sensor sites.
A2	Pipe roughness from design tables by material (clay 0.013, uPVC 0.010). The published roughness field is empty.	804 of 887 pipes are clay, and the data dates 308 of them to 1896. Old clay pipes with root intrusion, sediment or displaced joints run rougher and fuller than a clean design pipe, so the model may understate depths.	**High**	Measured depth and flow at one or two sites; CCTV condition grades.
A3	The construction year 1896 appears on 308 pipes.	This may be a real date or a default value used when the date is unknown. It only matters in the age-weighted leakage case, which gives these pipes the highest leakage weight.	**Low**	Ask SA Water whether 1896 is a placeholder.
A4	No energy losses at manholes, bends or junctions (no losses are set in SWMM).	Real chambers lose some head where pipes join or turn, which raises the upstream level. The model may understate depth at busy junctions on the trunk.	**Medium**	Sensitivity run with typical loss values.
A5	Every chamber is a 1,050 mm cylinder.	Affects only how much water a chamber stores. In settled runs this has almost no effect.	**Low**	None needed for steady runs.
A6	The three overflow (relief) pipes use their published inverts.	The settings file itself notes that one of them (4433446) then carries everyday flow: 70% of what leaves MH4451967 runs back to the N183 pump well, “a loop no real station has”. Levels near that station may be wrong.	**Medium**	Ask SA Water how these relief pipes are set; rerun with the “raised” option.

B. Sewage loads

No.	What the model does	Why it may differ from reality	Rating	How to check
B1	One peak factor of 2.0 for every pipe.	Peak factors are usually higher in small branches with few homes and lower on the trunk. The Harmon formula, which the code also contains, gives about 3.0 for the whole population of 9,700, so 2.0 may understate peak sewage. The Harmon case is in the full twelve-case set but not in the eight cases Sim 3 ran.	**High**	SA Water's own peaking rule; rerun with the Harmon case.
B2	Every connected property is treated as a home with 2.81 people using 200 L a day.	Shops, schools and other non-homes produce different amounts and at different times. Whether this over- or understates flow is not known.	**Medium**	Property type for each connection, or metered water use.
B3	Peak sewage and peak leakage are held constant together for up to 30 hours.	In reality the sewage peak lasts an hour or two, and leakage rises and falls with rain. Holding both at their peaks together is a simple "worst steady state". Whether it is higher or lower than real peaks is not known, because short storm peaks can be much larger than a constant rate.	**High**	A time-varying run with a daily pattern and a design storm.

No.	What the model does	Why it may differ from reality	Rating	How to check

C. Wet weather and outside flows

No.	What the model does	Why it may differ from reality	Rating	How to check
C1	Leakage (I&I) is a constant rate per 100 m of pipe, equal everywhere (one case weights it by pipe age).	Real leakage comes from particular defects, cracked laterals and illegal stormwater connections, so it is patchy. Patchy leakage changes which areas fill first, and so which manholes are chosen.	**High**	Wet-weather flow monitoring by area; smoke testing records.
C2	The four leakage rates (0, 0.11, 0.25, 0.40 L/s per 100 m) are assumed.	At Design wet, leakage (41.2 L/s) is already larger than sewage (33.4 L/s). With no growth, 7 manholes surcharge at Design wet, 107 of 328 at 0.25 and 141 at 0.40. If the real network does not surcharge this widely in wet weather, the higher rates are too high.	**High**	SA Water records of wet-weather surcharge or overflow in Walkerville.
C3	Flow from pipes outside the council data (25.5 L/s at Design wet, about a quarter of the total) is estimated from their length and the connection density of council pipes.	This is an estimate of homes and leakage not in the data. It is tested at $\times 1.5$ only; there is no lower case in the eight cases.	**Medium**	Flows at the 19 points where outside pipes join.
C4	The outlet at node 606 is a free outfall.	If SA Water's main beyond 606 runs full in wet weather, water backs up into Walkerville. The model would then understate depths near the outlet. A fixed-level option exists but was not used.	**Medium**	Level at or below node 606 in wet weather.

D. Pump stations

No.	What the model does	Why it may differ from reality	Rating	How to check
D1	The two pump stations (N183, N997) are SWMM “ideal” pumps, which pass on whatever arrives at once.	Real pumps have a fixed capacity and switch on and off. If inflow exceeds capacity, the well backs up. Pumping also arrives downstream in bursts, not as a steady flow. Levels upstream of the wells and at the discharge manholes may be understated. A rated-pump option exists but was not used in the Sim 3 runs.	**Medium**	Pump rates, well sizes and start and stop levels; SCADA run hours.

E. Growth

No.	What the model does	Why it may differ from reality	Rating	How to check
E1	Only one area grows in each run.	In reality growth happens across the suburb at the same time. The trunk then carries the sum of all areas, so its rise is larger than in any single-area run, and the trunk may reach the alarm or surcharge sooner than Sim 3 shows.	**High**	Whole-suburb growth runs (as done in Sim 3-e).
E2	New homes are spread over the area’s existing connections in proportion to the homes already there.	Infill tends to cluster where zoning allows it and where lots are large or old. A cluster puts more load on one or two branches than an even spread.	**Medium**	Council development approvals by street.
E3	New homes add sewage only, no leakage.	New connections and laterals add some leakage of their own, so growth may be slightly understated in wet cases.	**Low**	Leakage allowance per new connection from SA Water’s code.
E4	100% build-out is each area’s share of the PlanSA high case; the corridor takes the full 485 homes.	The projection is for the whole council, and the medium case is 390. If growth is slower or lands elsewhere, the build-out ladder reads at a different year.	**Medium**	Compare with the medium case and with approvals.

F. Detection and placement

No.	What the model does	Why it may differ from reality	Rating	How to check
F1	A sensor "sees" growth from the difference between two settled, noise-free depths.	A real sensor reads a level that changes through the day and from day to day, with its own measurement error. A 10 mm rise may be lost in that noise. Sim 3 handles this only with an assumed noise level in its time rule, so it may overstate how easily growth is seen.	**High**	A month of normal-day level records at one site.
F2	The 150 mm alarm is the same for every pipe size.	21 manholes are already above 150 mm in the Dry case with no growth, mainly on the larger trunk pipes. On those pipes the "alarm" rule is not a warning.	**Medium**	SA Water's own alarm rule by pipe size.
F3	A sensor set is judged by its worst case over the eight cases, each counted equally.	The cases are not equally likely. A placement tuned to the worst case may be weaker in the most common weather.	**Medium**	Weight cases by how often each occurs, once data allows.
F4	No blockages, partial blockages or root growth.	Tree roots are SA Water's main cause of blockages. A partial blockage raises the level upstream and could look like growth to a sensor, or hide it.	**Medium**	Blockage history by street; a blockage test case.

5. The biggest single limitation

The model has not been calibrated. No measured level or flow from the Walkerville network has been used to check any depth in Simulation 3. Most of the points above (roughness, leakage, peaking, outside flows, the outlet level) would be narrowed by one comparison against real data. Until then, Simulation 3 shows which manholes are *likely* to be useful sensor sites and why, not the exact depths at which they will read.